Academic Programs

PEOs, POs & PSOs

B.Tech ECE

Program Educational Objectives (PEOs)

The student in Electronics and Communication Engineering will be able to:

Produce Electronics and Communication Engineering Professionals with a solid foundation in Mathematics, Science and Technology which is essential to solve engineering problems.

Train students in good scientific and engineering practices so that they comprehend, analyze, design, and create novel products and offer solutions for industry specific processes and real life problems.

Prepare students to adopt the learning culture needed for a successful professional career by encouraging them to acquire higher qualifications, take up research and keep abreast of latest technological developments.

Inculcate organizing, managerial and entrepreneurship skills essential for professional growth.

Develop the consciousness among students towards moral values and professional ethics while developing innovative solutions to meet the societal needs.

Program Outcomes (POs)

Engineering Graduates will be able to:

Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences,and engineering sciences.

Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one?s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

Analyze, Design and Implement application specific Electronic systems for Analog and Digital domain, Communications, Signal and Image Processing Applications

Demonstrate the Computational and programming skills for problem solving

Identify and Apply Domain specific tools for Design, Analysis and Synthesis in the areas of VLSI and Embedded systems

M.Tech- VLSI System Design

Identify, formulate and analyze technical problems in areas like semiconductor technologies, VLSI Signal verification and Design verification and testing.

Design and implementation of VLSI architectures using FPGA

Use the techniques, skills, modern Electronic Design Automation(EDA) tools to evaluate and analyze the performance of the systems in VLSI domain.

Apply acquired knowledge from undergraduate engineering and other disciplines to identify, formulate and present solutions to technical problems related to various areas of VLSI.

Learn advanced technologies and analyze complex problems in the fields of VLSI.

Design and implementation of VLSI architectures using FPGA/CPLD.

Addressing specific problems in the field of VLSI system design in the form of mini projects, analysis and interpretation of data and synthesis of information to provide valid conclusions.

Use the techniques, skills, modern Electronic Design Automation (EDA) tools, software and equipment necessary to evaluate and analyze the systems in VLSI design environments.

Understand and commit to professional ethics, social responsibilities and norms of engineering practice.

Develop confidence for self-education and imbibe professional values for lifelong learning.

Demonstrate effective oral and written communication skills in accordance with technical standards.

Become knowledgeable about contemporary developments.

Ability to correct the mistakes effectively and learn from them to become good leaders.

Understand the scenario of global business.

Design and implementation of VLSI architectures using FPGA.

M.Tech- Embedded Systems

Demonstrate outstanding analytical & technical skills to evaluate , analyze and solve real time problems in Embedded Systems.

Apply the acquired knowledge to solve engineering problems to suit multidisciplinary situations.

Undertake research and development projects in the field of Embedded Systems.

Continue the personal development through professional study and self learning.

Demonstrate their professional, ethical and social responsibilities and contribute their part for addressing various global issues.

Apply the acquired knowledge from undergraduate engineering and other disciplines to identify, formulate and present solutions to technical problems related to various areas of Embedded Systems.

Learn advanced technologies and analyze complex problems in the fields of Embedded System design along with the fundamental concepts of engineering.

Addressing specific problems in the field of Embedded System in the form of mini projects, analysis, and interpretation of data and synthesis of information to provide valid conclusions by considering societal and environmental factors in the core areas of expertise.

Plan and conduct systematic study on a significant research topic effective to the societal, health, legal and environmental issues in the field of Embedded Systems.

Use the techniques, skills, modern Integrated Development Environment (IDE) tools, Operating systems, software and equipment necessary to evaluate and analyze the systems in Real time environments.

Ability to manage team effectively and become good leaders.

Develop confidence and motivation for self education and imbibe professional values for lifelong learning.

Become knowledgeable about contemporary developments by self learning.

Identify, formulate and solve engineering problems in the broad areas like System Design using Embedded Platforms, Applications in Signal Processing, Machine Vision and Communication Networks.

Use different software tools in the domain of Embedded Systems, for developing applications based on ARM processor core SoC and DSP processor.

Integrate multiple sub-systems to optimize performance of Embedded Systems and excel in industry sectors related to Embedded Systems domain.